

Development of a smart irrigation & sunshade system for chrysanthemum cultivation in greenhouse based on the internet of things

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Abstract. To be favored by nature in terms of climatic conditions, Dalat - Lam Dong of Vietnam has the leading agriculture in the country in terms of vegetable and flower products. One of the economically valuable flowers for Dalat agriculture is the chrysanthemum plant. For chrysanthemum strongly growth, it is necessary for monitoring and adjustment two important factors such as moisture and light. This study presents the process of design, implementation of the automatic monitoring, watering, and sun protection system based on humidity and light in greenhouses through the internet. The study combines the concept of the internet of things (IoT) and control systems. The control system uses micro-control STM32F103C8T6 as the central control to collect measurement data from the sensors and generate the control command for controlling the irrigation and sunshade system. Measurement data are sent to the Webserver by Lora protocol. The webserver interface is created by the IoT MyDvices Cayenne software which makes mobile devices accessible for convenient system monitoring. System test results work steadily with Lora communications within a range of 1.5 km and an average response time of 1s.

Keywords: Chrysanthemum cultivation, IoT technology, Lora gateway, Micro-processor, Irrigation & sunshade system, Humidity and sunlight measurement.

1 Introduction

In previous, despite having advantages in terms of climate, soil, and standing floriculture history, in general, the flower agriculture of Dalat city, Lam Dong province is still not better than other sections due to outdated farming techniques, fragmented planning, and small-scale cultivation mainly in households.

After 1994, the floriculture of Dalat city is more strongly developed when there was the appearance of foreign companies investing strongly with scientific farming techniques, the yield, and quality of flowers markedly increased. This has changed the traditional farming views of thousands of flower farmers in Dalat city [1]. Today, there are more and more individuals and businesses domestically and abroad investing in high-tech floriculture with revenue of many VND billions per year per hectare. This land is quickly becoming the agricultural capital of Vietnam's high-tech vegetables and flower farming, aiming to Dalat's agricultural products is reaching new heights to conquer the domestic market and export to foreign countries. Currently,

flower farming in Dalat city has grown tremendously in both quantity and quality by applying advanced technology in flower farming. However, the applications of automation technology, which was sensor technique not concerned due to high investment costs, the only foreign companies dare to make an investment, most individual flower growers still use human labor, farming efficiency is not high and still depends on nature. Nowadays, IoT technology has been widely used in agriculture cultivation for the measurement and monitoring of climate parameters. Moreover, it is even used for controlling such as the watering and light in the greenhouse [2-10].

The Chrysanthemum plant is sensitive to temperature, soil moisture, humidity, and light. To improve the chrysanthemum growth and its flower quality, it is necessary to automatically surveillance and adjusts these climate parameters in the best suitable. Several researchers [8-10] used IoT technology for the development of an automatic system for measurement, monitor, and control of the temperature, humidity, soil moisture, and light intensity in the greenhouse. In this research, the author wants to develop a smart irrigation & sunshade system that automatically maintained the moisture and light in the greenhouse fall in their optimal value ranges. If one of these parameters is out of its threshold, the proposed system will automatically work for tuning these parameters goes in their threshold values.

2 Material and method

2.1 Systematic procedure for developing a proposed system

To develop the proposed system, the chrysanthemum biology is first analyzed to find the optimal climate parameters in the greenhouse for strong chrysanthemum growth, the greenhouse structure is then designed, and the irrigation & sunshade system is also designed. Based on the analysis of the system operation mechanism, a control system is designed for controlling the irrigation and sunshade operation which to adjust the humidity and light parameters to reach the desired values. After that, the wireless communication is designed using the Lora device, the climate parameters in the greenhouse can be monitored by mobile devices through a web browser. The web-server interface is developed using IoT Cayenne software. A systematic procedure for developing a proposed system is shown in Fig.1.

2.2 Chrysanthemum biology analysis

Chrysanthemums grow optimally at an altitude of 700-1200m above sea level. It is a short-day flowering plants, the growth period of the plant is from 10-12 weeks, including three stages: the root development phase (1-2 weeks), the plant growth phase (from 6-8 weeks), and the period from bud break to harvest (2-3 weeks). During the growth of chrysanthemums, there are harsh requirements on climate conditions, on the one hand, the plant needs water but cannot withstand rain, so it is necessary to design an appropriate irrigation system. During the flowering period, chrysanthemum plants are also not tolerant of direct sunlight and need to be treated with light to shorten the day to promote the differentiation of flower buds and adjust the flowering time

appropriately. Based on the chrysanthemum technology guide of Lam Dong agriculture extension center [11], suitable climate parameters for chrysanthemum cultivation in periods are recommended in Table 1.

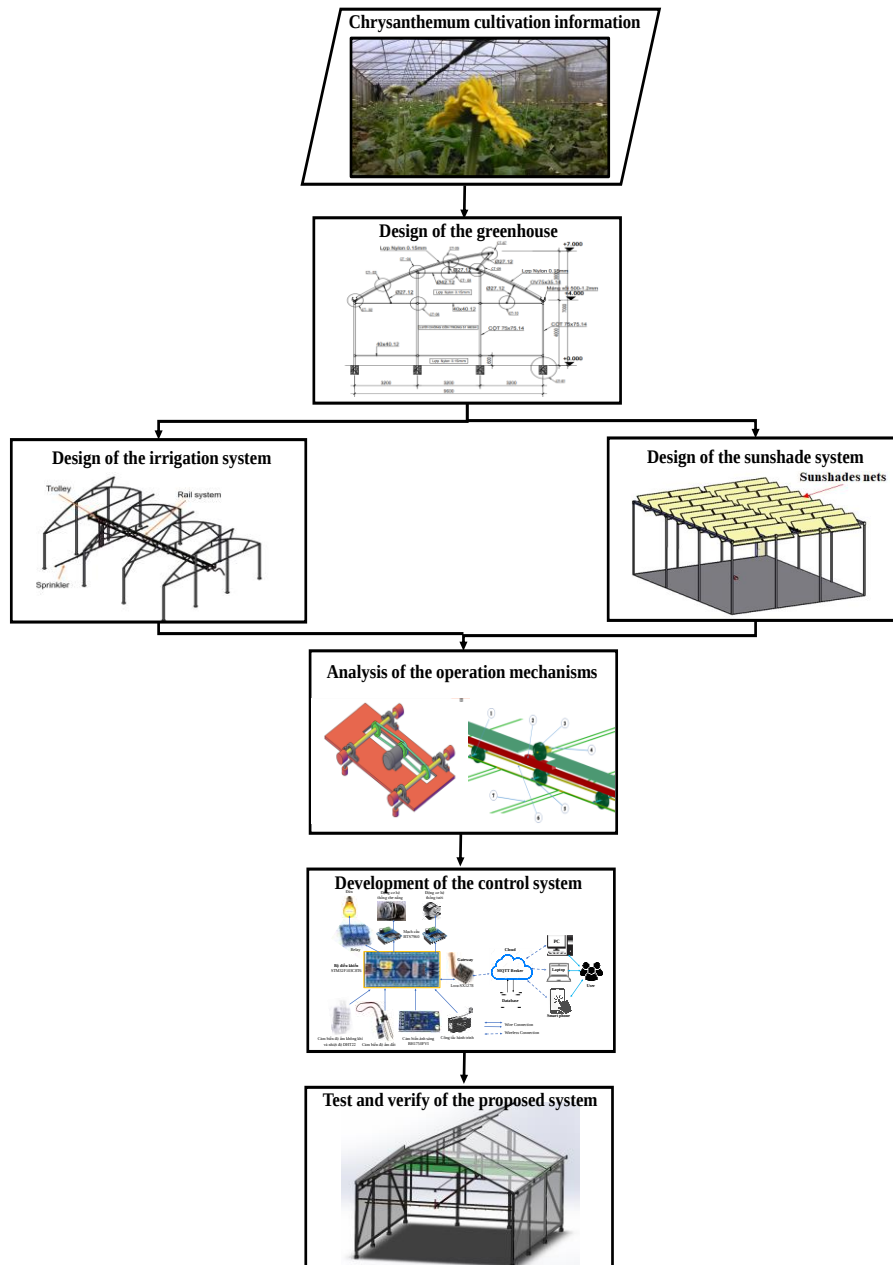


Fig. 1. Systematic procedure for developing a proposed system.

Table 1. Recommended climate parameters for chrysanthemum growth in greenhouse.

Chrysanthemum growth stages	Time	Parameters	Units	Accepted thresholds
Roots	1-2 weeks	Temperature	°C	22÷26°C
		Humidity	%	90÷95%
		Soil moisture	%	70÷80%
		Light intensity	h (hour)	-
Trunks	6-8 weeks	Temperature	°C	15÷20°C
		Humidity	%	65÷70%
		Soil moisture	%	70÷80%
		Light intensity (32-108 Lux)	h (hour)	6÷8h
Blooming	2-3 weeks	Temperature		16÷18°C
		Humidity		65÷75%
		Soil moisture		70÷80%
		Light intensity (32-108 Lux)		7÷9h

2.3 Mechanical system design

The irrigation system design

To maintain the climate parameters in the greenhouse such as temperature, soil moisture, and air humidity are in the selected ranges, it is necessary to design the automatic irrigation system. There are three famous irrigation system types: localized irrigation, drip irrigation, and sprinkler irrigation [12]. The sprinkler irrigation system is considered a suitable solution for the shrub plant as the chrysanthemum by forming the dewdrops spray onto the surface of the plants. The sprinkler irrigation method uses the water hose and nozzles which are hinged on the high and horizontal movements to provide enough water to plants in a large area. The design of the sprinkler irrigation system is shown in Fig.2.

The nozzles of the sprinkler irrigation are mounted on the bracket which is fixed in the trolley and movement along with the trolley. The water hose which supplies water to sprinklers is hanged on cable and it is extended or contracted when the trolley moves forward or reverse. The trolley is driven by a DC motor through the belt transmission, it moves on the rails based on four wheels, four guide wheels are mounted to contact the inside of the rails to protect the trolley is not derailed out of the rails. The irrigation system operation is automatically controlled by the main controller when the temperature or soil moisture is out of their thresholds. Two the ends of the rails are installed two limit switches and the dampers aiming to reverse the trolley direction.

The Sunshade system design

The sunshade net is used to block a part of the sunlight to reach the ground and creates shade for plants as well as to prevent the harmful ultraviolet rays by UV coating

on the surface of the material. In addition, the sunshade net also helps to limit strong winds that can affect weak trees and reduce the force of raindrops on the ground to avoid scouring and soil erosion during heavy rain. The designed sunshade system is shown in Fig.3. It consists of the multi-sunshade pieces which have the dimension are 2000x1000 and are installed above the irrigation system. The sunshade pieces are fixed in the gears and they are driven by the DC motor (4) through the rack-and-gear transmission (2,3). The adjustment of sunlight in the greenhouse is executed by turning the gear (3).

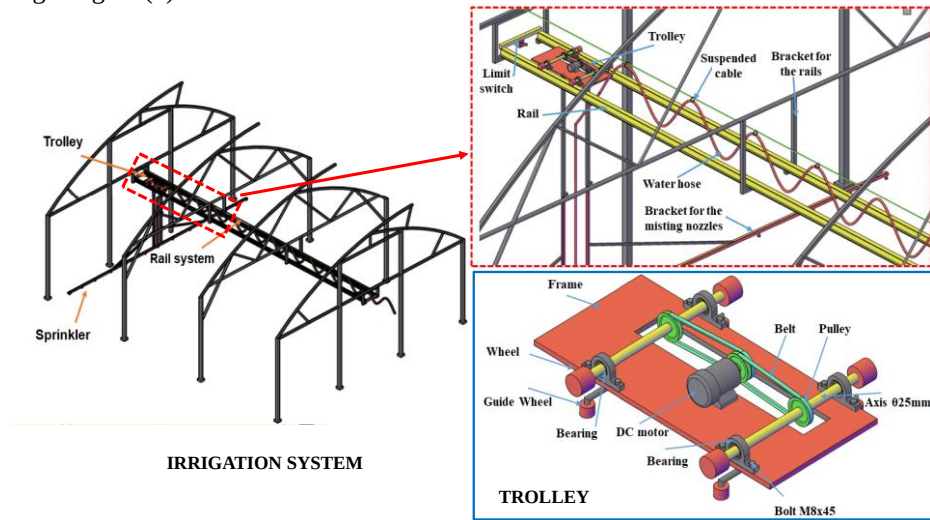


Fig. 2. The sprinkler irrigation system.

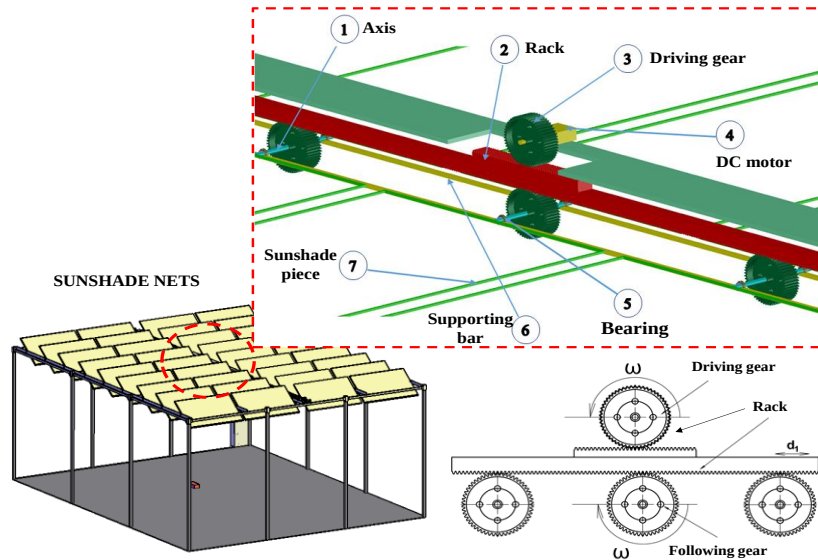


Fig. 3. The sunshade system.

2.4 Control system design

Hardware design

The control system consists of five main units such as the central controller, the sensors, the actuators, the wireless communication, and the webserver interface. The block diagram of the proposed control system for the chrysanthemum cultivation in the greenhouse is shown in Figure 4.

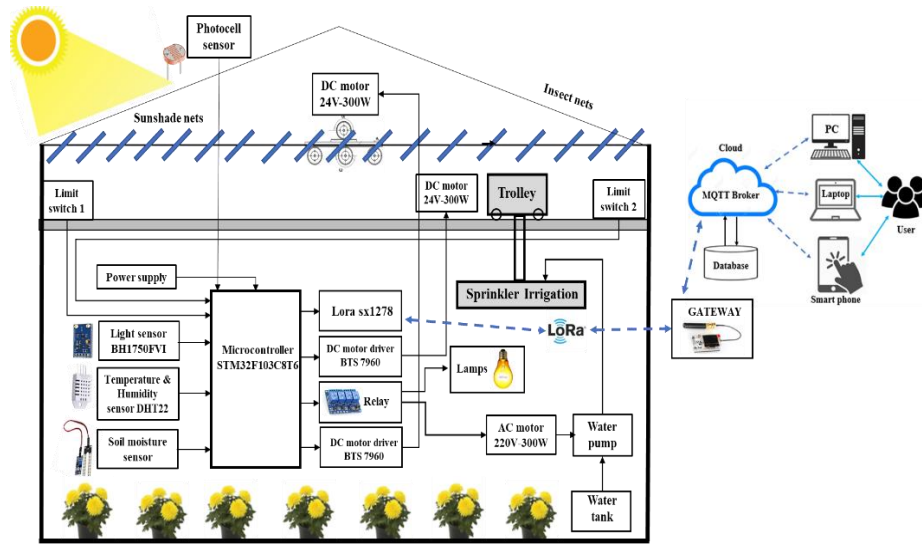


Fig. 4. The block diagram of the proposed control system.

Software design

In this project, three types of programming are used, namely: programming in a microcontroller, programming in a gateway, and programming in a web interface. The microcontroller program is made in STM32F103C8T6 for measurement of the climate parameters, controlling the irrigation and sunshade operations, and send measured data to the gateway through the Lora SX1278 module. The gateway program is developed for receiving data packages from Node (STM32F), storage to the database, and send to the webserver, conversely, receiving commands from the web server, comparison measured values with their thresholds, creating the data packages, and sending the packages to Node through Lora protocol. The flowcharts of Node are shown in Fig.5.

The web-server interface program is created using the Cayenne myDevices software. The Cayenne is an IoT platform with superior features such as data visualization, easy to use by drag-and-drop, the dashboard is created to visualize data coming from remotely connected devices or control them, using widgets to visualize information, supporting several devices (Arduino, Raspberry, ESP,...). Flow chart gateway and the dashboard created Cayenne software are shown in Fig.6.

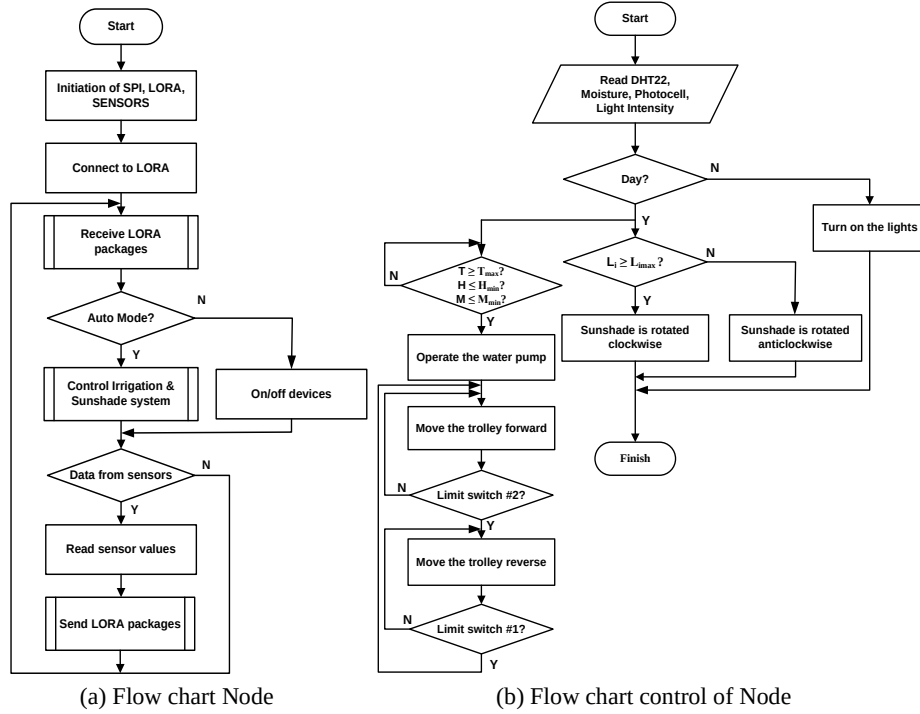


Fig. 5. Flow chart of Node in Lora network.

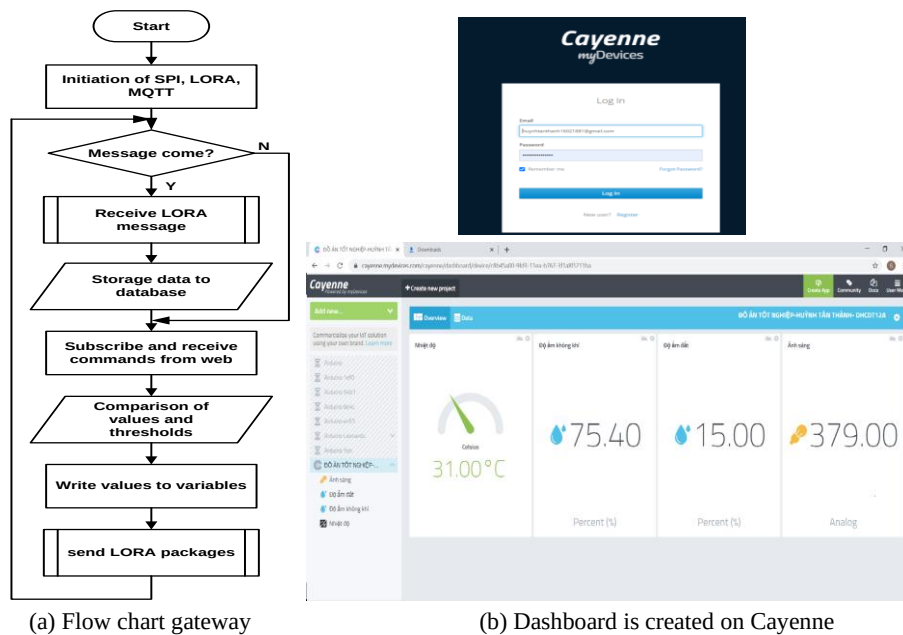


Fig. 6. Flow chart gateway and web-interface

3 Results and conclusion

3.1 Implementation

Before applying the proposed solution for the reality of the chrysanthemum greenhouse, the proposed method will be implemented and verification on the testbed. The testbed was built and shown in Fig.7. The power supply of the devices including a 12VDC for two DC motors, and a voltage 5VDC for others such as sensors, micro-controller, ESP32 Lora kit, bridge circuit BTS7960. Pin mapping between microcontroller and devices is described in Table 2.

Table 2. Pin Mapping.

Controller	PIN	Devices
STM32F103C8T6 BLUE PILL	PB_8 (I2C) - 2(Data)	DHT22
	PB_3 - analog in	Soil Moisture
	PA_0, PA_1 - SDA, SCL	Light Intensity
	PB_4 – analog in	Photocell
	B1, B0, B11 – R_PWM, L_PWM, R_EN&L_EN	BTS7960_1
	B6, B7, B10 – R_PWM, L_PWM, R_EN&L_EN	BTS7960_2
	A5, A7, A6, A12, A11, B15, B14-SCK, MISO,	Lora SX1778
	MOSI, RESET, DI00, DI01, DI02	

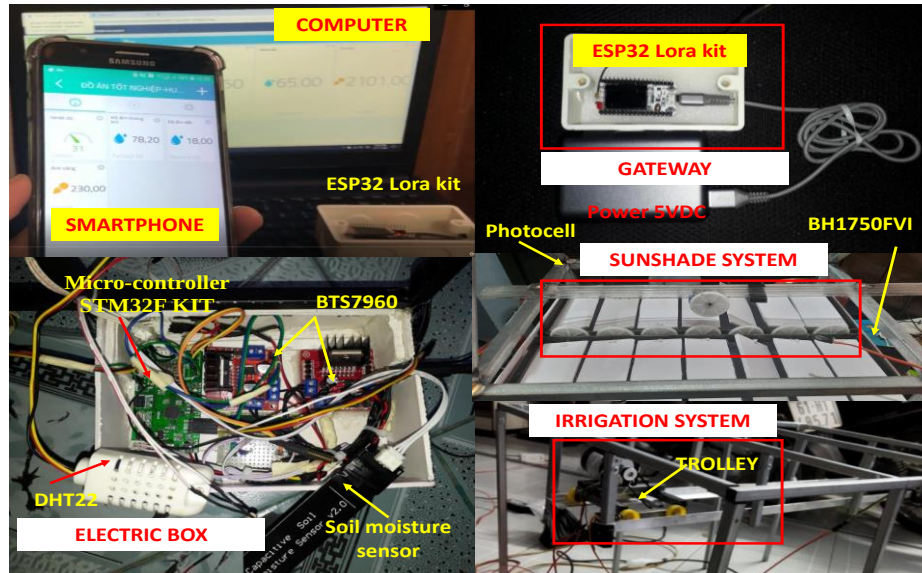


Fig. 7. Hardware implementation

3.2 Testing results

After the devices are mounted in the testbed, wiring connection and parameters setting are then executed. The system testing is executed with the distance of the gateway over 1km from the testbed. The measured values are also storage the database. The monitoring of the measured results is shown in Fig.8. The control tasks satisfy the desired targets with time response below 1 second.

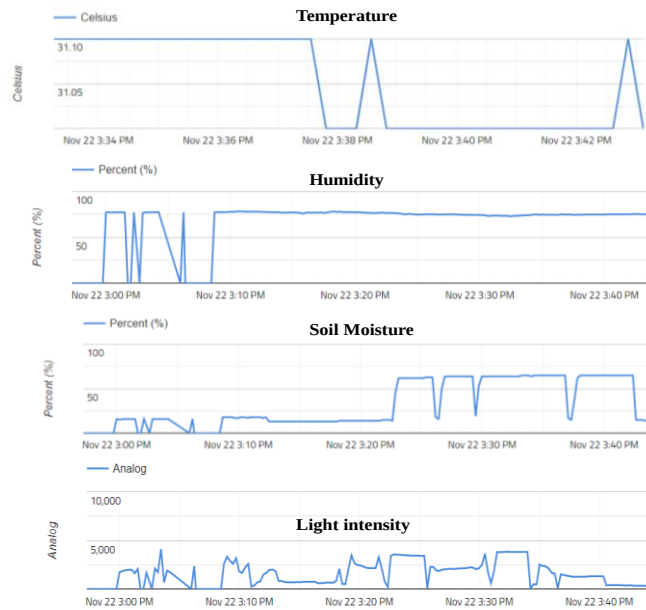


Fig. 8. Monitor of the measured climate parameters results.

In order to verify the measured results of the developed system. These measured results can be compared with government environment data. Users can view the environment data by access in address <http://vnweather.net/>.

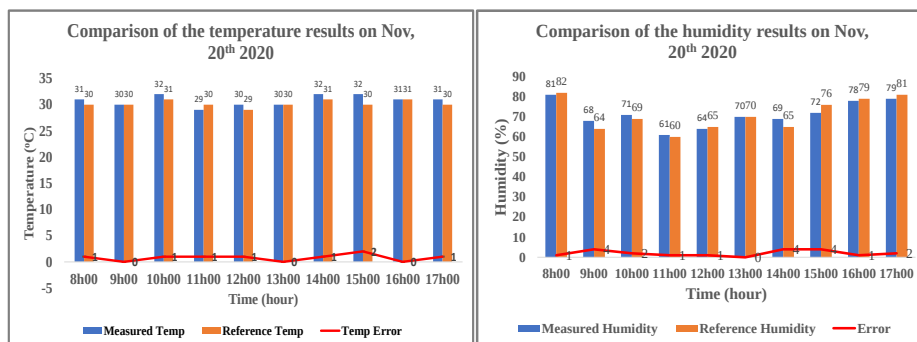


Fig. 9. Monitor of the measured climate parameters results.

3.3 Conclusion

Based on the testing results of the proposed system. It can be concluded that this proposed method achieves these the following benefits:

- Real-time measurement and monitoring the climate parameters in the greenhouse with high accuracy.
- Automatic sprinkler irrigation and sunshade system based on the changes the measured parameters compared with their threshold values. Timely response to control task below 1 second.
- Sensor data are stored in the database up to one year
- Monitoring and controlling the system can be accessed by various types of smartphones, as well as browsers.

The achieved experiment results on testbed demonstrated that the proposed solution can able to apply the real greenhouse in Dalat city.

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